

Union Exercise 4

Question 1

```
( SELECT Account_ID,  
        Account_Holder,  
        (Johannesburg) 'Sandlen' AS City,  
FROM branch - Sandlen - accounts )
```

Union

```
( SELECT Account_ID,  
        Account_Holder,  
        (Johannesburg) 'Rosebank' AS City,  
FROM branch - Rosebank - accounts ; )
```

Account ID	Account - Nam	City
A001	Morwale Dlamini	Sandlen Johannesburg
A002	Daniel Mokueng	Sandlen Johannesburg
A003	Lerato Silubhe	Sandlen Johannesburg
A004	Pete Nkosi	Sandlen Johannesburg
A005	Zanele Khumalo	Rosebank Johannesburg
A006	Thabo Mutha	Rosebank Johannesburg

Question 2

```
( SELECT Product_Code,  
        Product_Name,  
        Product_Code AS Saving )
```

```
FROM Saving_Product  
UNION
```

```
SELECT Product_Code  
        Product_Name
```

```
Product_Code AS current  
FROM current_products
```

Product_Code

Product_Name

Question 2

```
( SELECT Product_Code,
```

```
    CASE Product_Name,
```

```
    WHEN 'Product_Code' = 'SV' THEN 'Saving',
```

```
    WHEN 'Product_Code' = 'CR' THEN 'Current'
```

```
END AS Product_Type
```

```
FROM Saving_Product )
```

UNION

```
( SELECT Product_Code,
```

```
    Product_Name,
```

CASE

```
    WHEN Product_Code = 'SV' THEN 'Saving',
```

```
    WHEN Product_Code = 'CR' THEN 'Current'
```

```
END AS Product_Type
```

```
FROM Current_Product ; )
```

Product Code	Product Name	Product type
SV01	Basic Saving	Saving
SV02	Premium Saving	Saving
SV03	Youth Saving	Saving
SV04	Business Saving	Saving
CR01	Standard Current	Current
CR02	Gold Current	Current
CR03	Business current	Current

Question 3

(SELECT Staff_ID,
Staff_Name,
Replace Staff_Name with 'Email'
from retail-banking-staff)
Union

(SELECT Staff_ID,
Staff_Name,
Replace Staff_Name with 'email'
from Corporate-banking-staff
order by Staff_ID)

Staff_ID	Staff_Name	email
S001	Mpho Radebe	mpho.radebe@brightbank.co.za
S002	Brian Tshabalala	Brian.Tshabalala@brightbank.co.za
S003	Aisha Patel	AishaPatel@brightbank.co.za
S004	Kabelo Makhobe	Kabelo.M@brightbank.co.za
S005	Mantle Dube	MantleDube@brightbank.co.za
S006	Sipho Khumalo	Sipho.K@brightbank.co.za

Question 4

SELECT City-Code,
City-Name,

CASE city_code

WHEN 'C01' THEN 'Gauteng'

WHEN 'C02' THEN 'Gauteng'

WHEN 'C03' THEN 'Western Cape'

WHEN 'C04' THEN 'KwaZulu-Natal'

WHEN 'C05' THEN 'Limpopo'

WHEN 'C06' THEN 'Eastern Cape'

END AS region

FROM mobile-branch-cities

UNION

SELECT City-Code,
City-Name,

CASE

WHEN 'C01' THEN 'Gauteng'

WHEN 'C02' THEN 'Gauteng'

WHEN 'C03' THEN 'Western Cape'

WHEN 'C04' THEN 'KwaZulu-Natal'

WHEN 'C05' THEN 'Limpopo'

WHEN 'C06' THEN 'Eastern Cape'

END AS region

FROM digital-branch-cities;

City-Code	City-Name	Region
C01	Johannesburg	Gauteng
C02	Pretoria	Gauteng
C03	Cape-town	Western-Cape
C04	Durban	KwaZulu-Natal
C05	Pulokwane	Limpopo
C06	Port Elizabeth	Eastern Cape

Question 6

```
( SELECT txn_id AS transaction_id,  
      Account_ID,  
      amounts,  
      Null AS transaction_date ... not available in  
FROM atm01_transactions )
```

Union All

```
( SELECT txn_id AS transaction_id,  
      Account_ID,  
      Amounts,  
      Null AS transaction_date ... not available in  
FROM atm02_transactions; )
```

transaction_id	account_id	amount	transaction_date
T1001	A001	500	Null
T1002	A002	1200	Null
T1003	A001	300	Null
T1003	A002	300	Null
T1004	A003	250	Null
T1005	A004	900	Null
T1006	A002	450	Null
T1007	A005	50	Null

Question 7

(SELECT app_id AS application_id,
customer_id,
loan_type,

Null AS amount_requested -- Not available in source tables
FROM grantery_loan_applications)

Union All

(SELECT app_id AS application_id,
customer_id,
loan_type,

Null AS amount_requested -- Not available in source tables
FROM western_type_loan_applications;

Application_id	Customer_id	loan_type	Amount_requested
LA001	C1001	Home loan	Null
LA002	C1002	Vehicle loan	Null
LA003	C1003	Personal	Null
LA003	C1003	Personal	Null
LA004	C1004	Home loan	Null
LA005	C1005	Vehicle loan	Null
LA006	C1006	Business	Null
LA007	C1007	Home loan	Null

Question 8

```
( SELECT Complaint- id  
      Customer- id  
      Null As Category,  
      Null As Logged- date.  
FROM email- complaints )
```

Union All

```
SELECT Complaints- id,  
      Customer- id,  
      Null As Categories,  
      Null As Logged- date  
FROM app- complaints  
order By Complaints- id
```

Complaints- id	Customer- id	Category	logged- date
AC001	C2005	Null	Null
AC002	C2001	Null	Null
AC003	C2006	Null	Null
AC004	C2007	Null	Null
EC001	C2001	Null	Null
EC002	C2002	Null	Null
EC003	C2003	Null	Null
EC004	C2004	Null	Null

Question 9

```
SELECT Payment_Id,  
        Account_Id,  
        Amount,  
        Null AS payment_date -- Not available in source table  
FROM April-payments
```

Union All

```
SELECT payment_id,  
        Accounts_id  
        Amount,  
        Null AS payment_date  
FROM May-payments  
Order By payment_id
```

Payment_Id	account_id	amount	payment_date
Pay 001	Acc1	1250	Null
Pay 002	Acc2	4500	Null
Pay 003	Acc3	4200	Null
Payment 4	Acc4	3300 12500	Null
Payment 5	Acc4	12600	Null
Payment 6	Acc5	7600	Null
Payment 7	Acc2	5100	Null
Payment 8	Acc6	2800	Null

Question 10

SELECT entry_id,
Account_id,
entry_type,
Null AS amount,
Null AS entry_date
FROM debit_entries

Union All

Select entry_id,
Account_id,
entry_type,
Null AS amount,
Null AS entry_date
FROM debit_entries ;
order by entry_id.

entry_id	account_id	entry_type	amount	entry_date
CR001	ACC1	Credit	Null	Null
CR002	ACC5	Credit	Null	Null
CR003	ACC2	Credit	Null	Null
CR004	ACC6	Credit	Null	Null
DR001	ACC1	Debit	Null	Null
DR002	ACC2	Debit	Null	Null
DR003	ACC3	Debit	Null	Null
DR004	ACC4	Debit	Null	Null

Bonus Question

Question 1

- ① Use Union $\rightarrow$ The objective is to have a unique list of a customer who shows up on both the Jan and Feb results but should only appear once in the results.
Union removes duplicate rows across.

Question 2

- Use Union All $\rightarrow$ The goal is to count every individual transaction & records, including intentional or system driven repeats.
Union All repeats transactions they duplicate transactions as they want to account for every transaction.

Question 3

- The Query fails - because the two tables don't have the same number of columns.

There is additional column named City which is not available in table 1.

This violates one of the core rules of Union.

Every Select Statement being combined must return the same number of columns, in the same order, with compatible data types.